

Project Scope & Proposal

Centralised Data Platform — Phase 1: Property Valuation & Market Intelligence

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Executive Summary

The original brief describes a full centralised data platform covering five services: a listings marketplace, property management, market insights, smart search, and mortgage. After reviewing the data currently available — a single dataset of 2,090 property records with sale and assessment details — I recommend we narrow the scope of the three-week delivery to what this data actually supports, and position the work as **Phase 1 of a multi-phase platform** rather than attempting to deliver all five services at once.

Phase 1 will deliver an **Automated Valuation Model (AVM)**, a **Market Intelligence analytics layer**, and an **Assessment Ratio flag** — all grounded in the dataset we have today. The remaining services (marketplace listings, property management, mortgage) are held back for later phases that require additional data sources to be onboarded.

Why We Are Narrowing the Scope

The dataset AlloyTower has provided is a **property-tax-and-assessment snapshot**: each of the 2,090 rows describes a property's physical attributes (bedrooms, bathrooms, sqft, lot size, year built), its most recent recorded sale, and its county assessor record. This is a strong foundation for valuation and market analysis, but it does not contain the data needed to deliver several of the services in the original brief.

Known Data Gaps

The following gaps exist in the current dataset. These are constraints Phase 1 is designed to work around, and they define what additional data would need to be onboarded for Phase 2 to unlock the remaining services in the original brief.

- **No active-listing data.** The dataset contains recorded sales and assessments, not live listings. There is no list price, listing status, days-on-market, or listing agent — so a marketplace service cannot be built from this data alone.
- **No user or behavioural data.** There is no record of who viewed, saved, or inquired about which properties. Personalised recommendations are not possible; only item-to-item similarity is.
- **No tenant, lease, or rent-payment records.** The property-management service cannot be built without these.
- **No borrower or loan data.** The mortgage service cannot be built without applicant income, credit, and loan-term data.
- **Multi-year sale prices are mixed together.** The dataset includes sales from 2017 to 2025 at their original sale-date prices, which mixes 2017 dollars with 2025 dollars. Phase 1 will handle this by

including sale date as a model feature, but a true real-time market view would need a continuously refreshed feed.

- **Single static snapshot.** There is no ingestion pipeline behind the dataset — it is a one-off extract. Any claim of "real-time market insights" would require a live data source to be connected in a later phase.
- **No rental price data.** Only sale prices are present, so the valuation model predicts sale value, not rental value. A rental AVM would require separate rental data.

Data coverage against the five core services

The table below maps each service in the brief against the data available today. Only two of the five services have meaningful data coverage.

Service	Data needed vs. available	Coverage	In scope?
Marketplace	Needs listing status, list price, listing date, agent, photos. None present.	None	No — Phase 2
Property Management	Needs tenant records, leases, rent ledger, inquiry logs, maintenance tickets. None present.	None	No — Phase 2
Market Insights & Analytics	Historical sale prices, price-per-sqft, and property attributes are all present. Aggregation by city, zip, and property type is viable.	Partial	Yes — Phase 1
Smart Property Search	Property attributes (item side) are rich. User behaviour data (searches, saves, clicks) is absent, so personalisation is not possible — only item similarity.	Item side only	Optional stretch
AlloyTower Mortgage	Needs borrower income, credit history, loan terms. None present.	None	No — Phase 2

The risk of not narrowing scope

Attempting to deliver all five services in three weeks would require fabricating three of them from data we do not have. The resulting platform would demo badly, mislead stakeholders about real capability, and create technical debt when real data is onboarded later. Scoping to Phase 1 lets us ship something defensible and honest within the timeline and sets up Phase 2 cleanly.

What Phase 1 Will Deliver

1. Automated Valuation Model (AVM) — the headline deliverable

Target: *last_sale_price* — the recorded sale price of a property.

What it does: Given a property's attributes (location, size, age, type, bedrooms, bathrooms, lot size), the model predicts its market value. This is the same class of model behind Zillow's Zestimate, Redfin Estimate, and HouseCanary — the industry-standard approach to digital property valuation.

Why it matters: The brief cites that 67% of real estate transactions are delayed due to poor data quality and fragmented systems. Valuation uncertainty is a major driver of that delay. An AVM replaces manual appraisal guesswork with a consistent, data-driven estimate that refreshes whenever property attributes or market conditions change.

Approach: Gradient-boosted trees (XGBoost or LightGBM) as the primary model, with linear regression as a baseline to demonstrate lift. Sale date will be included as a feature to account for the fact that the dataset spans multi-year sales (2017–2025).

Success metrics: We will measure the model using **Mean Absolute Percentage Error (MAPE)** — a way of asking "on average, how far off are the model's price guesses, as a percentage?" If a house actually sold for \$500,000 and the model guessed \$450,000, it was off by \$50,000, which is 10% of the real price. MAPE is the average of that percentage across every property the model predicts. Lower is better: a MAPE of 10% means the model is typically within 10% of the right answer; 5% means typically within 5%.

Industry benchmarks for AVMs are roughly 2–7% MAPE on active listings and around 10% on off-market properties. Given the size of this dataset, we will target under 15% MAPE, with a stretch goal of 10%. We will also report **Mean Absolute Error (MAE)** — the same idea but expressed in dollars rather than percentages — so stakeholders can see the typical miss in real money.

2. Market Intelligence Analytics Layer

A set of aggregated views and visualisations that surface pricing trends and demand signals from the dataset. This is the BA- and stakeholder-facing deliverable and directly addresses the "Market Insights & Analytics" service in the brief.

Included views:

- **Price-per-sqft heatmaps** by city, state, and property type.
- **Sale trends over time** showing how prices have moved year-on-year across major metros.
- **Inventory composition** — breakdown of property types, age bands, and size distributions by location.
- **Assessment vs. market** — average gap between assessed values and actual sale prices, by jurisdiction.

Can be covered by Business Analysts and Data Analysts.

3. Assessment Ratio Flag (secondary insight)

Target: Ratio of *assessed_value* to *last_sale_price*, and a binary flag for properties significantly under- or over-assessed.

Use case: supports property-tax appeal workflows and investor screening (identifying undervalued assets). This is essentially free given the columns we have and adds a concrete analytical deliverable beyond the AVM.

4. (Optional stretch) Similar-Properties Lookup

A content-based item-similarity model — given a property, return the N most similar properties by attribute. Powers a "you might also like" experience without requiring any user behaviour data. Will only be built if Phases 1–3 complete ahead of schedule.

Explicitly Out of Scope for Phase 1 (Our 3-sweek Project)

Flagging these now so expectations are aligned from the start:

- Live listings marketplace (no listings data available).
- Tenant, lease, or rent-payment management (no tenant or lease data available).
- Mortgage pre-approval and loan offers (no borrower or loan data available).
- Personalised recommendations based on user behaviour (no user event data available).
- Real-time data ingestion pipelines — Phase 1 works against a static snapshot.

Recommended Phase 2 (for discussion)

To unlock the remaining services from the original brief, the following data sources would need to be onboarded in a later phase:

- **Listings feed** — active-listing data including list price, days-on-market, and status, to enable the marketplace and demand analytics.
- **Inquiry and click logs** — user event streams from the website or app, to enable personalised search.
- **Tenant and lease records** — to enable the property-management service.
- **Loan application data** — to enable the mortgage service.

If any of these already exist in AlloyTower's systems and were simply not included in the Phase 1 dataset, I would recommend a short discovery session before Phase 1 ends to scope Phase 2 properly.

Fit Against the Three-Week Plan

This scope maps cleanly onto the three-phase structure in the original brief:

Phase	Deliverables aligned to this scope
Week 1 — Design & Development	Finalise BRD and data dictionary Exploratory data analysis and data-quality assessment Feature engineering and AVM model design
Week 2 — Go-Live & Deployment	AVM trained, evaluated, and optimised Assessment ratio flag implemented Analytics dashboard finalised and validated UAT with BA and stakeholders

Phase	Deliverables aligned to this scope
Week 3 — Documentation & Presentation	Model documentation and methodology write-up Lessons learned and Phase 2 recommendations Final stakeholder presentation

Summary

Phase 1 delivers a defensible, data-grounded slice of the centralised platform: an industry-standard valuation model, a market intelligence layer, and an assessment insight — all built from the data we have. It avoids overcommitting on services the current dataset cannot support, and it sets up a clear Phase 2 conversation once additional data sources are available.